

Project Proposal

Group 15

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Abstract

Using financial ratios and available market data, the team would like to understand if a logistic regression model can be built such that members of the S&P500 index can be identified prior to realizing outsized returns, when compared to the total index performance. Any equity that has a return percentage greater than the total benchmark percentage return for the same time period will be classified as an 'outperformer', while those that had less than the market return will be classified as 'underperformers'. After training a logistic regression model, and hopefully finding statistical significance, the team will then apply the results to an ensuing time period's benchmarks, to determine portfolio performance of 'outperformers' against the standard benchmark. Class imbalances will also need to be accounted for, as a 50/50 split for underperformer and outperformer is unlikely.

The preliminary dataset the team has collected has yearly information, but the team is actively searching for quarterly data as well for the same analysis, and may choose to slice the data into quarters for analysis.

The team shall also encompass other data analysis techniques including Linear Regression (to establish and correct for the effects of multicollinearity (perhaps using VIF scores). Moreover, the team may look to see the significance of clustering as well - to determine if companies within the same sector have similar ratios and/or if slicing the data across market capitalization parameters has an affect.

Data Sources

Potential data sources include:

- Yahoo Finance
- Wharton Research Data Services
- SimFin

Inspiration

U. Ananthakumar and R. Sarkar, "Application of Logistic Regression in Assessing Stock Performances," 2017 IEEE 15th Intl Conf on Dependable, Autonomic and Secure Computing, 15th Intl Conf on Pervasive Intelligence and Computing, 3rd Intl Conf on Big Data Intelligence and Computing and Cyber Science and Technology Congress(DASC/PiCom/DataCom/CyberSciTech), Orlando, FL, USA, 2017, pp. 1242-1247, doi: 10.1109/DASC-PiCom-DataCom-CyberSciTec.2017.199